

Y2 Electromagnetism 2

Non-assessed problem solutions 5 (weeks 9–11)

EM waves in dielectrics, conductors and plasma



1) The opening angle is

$$\theta_0 = \begin{cases} 2 \arcsin \sqrt{n^2 - 1} & \text{if } 1 \leq n < \sqrt{2}, \\ \pi & \text{if } n \geq \sqrt{2}. \end{cases}$$

2) Hint: use the expressions for Brewster's angle and Snell's law of refraction.

3) The refractive index is

$$n = 1 / \tan \frac{A}{2} = 1 / \tan 30^\circ = \sqrt{3} = 1.73.$$

4) The degrees of polarisation are

$$\begin{aligned} \Delta_R &= 1, \\ \Delta_T &= \frac{4n^2 - (n^2 + 1)^2}{4n^2 + (n^2 + 1)^2} = -0.08. \end{aligned}$$

5) Use the equation of motion of electrons, Ohm's law $\vec{j} = \sigma \vec{E}$, and the expression for the electron drift speed.
